

ADS EXP 1: Explore the descriptive statistics on the given dataset

```
import pandas as pd
import numpy as np

df = pd.read_csv('supermarket_sales - Sheet1.csv')
df.head()

{"columns":[{"name":"index","rawType":"int64","type":"integer"},
{"name":"Invoice ID","rawType":"object","type":"string"},
{"name":"Branch","rawType":"object","type":"string"},
{"name":"City","rawType":"object","type":"string"},{"name":"Customer
type","rawType":"object","type":"string"},
{"name":"Gender","rawType":"object","type":"string"},{"name":"Product
line","rawType":"object","type":"string"},{"name":"Unit
price","rawType":"float64","type":"float"},
{"name":"Quantity","rawType":"int64","type":"integer"},{"name":"Tax
5%","rawType":"float64","type":"float"},
{"name":"Total","rawType":"float64","type":"float"},
{"name":"Date","rawType":"object","type":"string"},
{"name":"Time","rawType":"object","type":"string"},
{"name":"Payment","rawType":"object","type":"string"},
{"name":"cogs","rawType":"float64","type":"float"},{"name":"gross
margin percentage","rawType":"float64","type":"float"},{"name":"gross
income","rawType":"float64","type":"float"},
{"name":"Rating","rawType":"float64","type":"float"}],"ref":"ac7f43c0-
dcfe-4a7b-92e5-6a3931a24efd","rows":[["0","750-67-
8428","A","Yangon","Member","Female","Health and
beauty","74.69","7","26.1415","548.9715","1/5/2019","13:08","Ewallet",
"522.83","4.761904762","26.1415","9.1"],["1","226-31-
3081","C","Naypyitaw","Normal","Female","Electronic
accessories","15.28","5","3.82","80.22","3/8/2019","10:29","Cash","76.
4","4.761904762","3.82","9.6"],["2","631-41-
3108","A","Yangon","Normal","Male","Home and
lifestyle","46.33","7","16.2155","340.5255","3/3/2019","13:23","Credit
card","324.31","4.761904762","16.2155","7.4"],["3","123-19-
1176","A","Yangon","Member","Male","Health and
beauty","58.22","8","23.288","489.048","1/27/2019","20:33","Ewallet",
"465.76","4.761904762","23.288","8.4"],["4","373-73-
7910","A","Yangon","Normal","Male","Sports and
travel","86.31","7","30.2085","634.3785","2/8/2019","10:37","Ewallet",
"604.17","4.761904762","30.2085","5.3"]],"shape":
{"columns":17,"rows":5}}

#Mean SD LQ UQ max min
df.describe()

{"columns":[{"name":"index","rawType":"object","type":"string"},
{"name":"Unit price","rawType":"float64","type":"float"},
{"name":"Quantity","rawType":"float64","type":"float"},{"name":"Tax
```

```

5%","rawType":"float64","type":"float"},
{"name":"Total","rawType":"float64","type":"float"},
{"name":"cogs","rawType":"float64","type":"float"},{"name":"gross
margin percentage","rawType":"float64","type":"float"},{"name":"gross
income","rawType":"float64","type":"float"},
{"name":"Rating","rawType":"float64","type":"float"}],"ref":"c8b3df3a-
3d07-4058-84a4-a6b72a8a7e74","rows":
[["count","1000.0","1000.0","1000.0","1000.0","1000.0","1000.0","1000.
0","1000.0"],
["mean","55.67213","5.51","15.379368999999999","322.966749","307.58738
","4.761904762","15.379368999999999","6.9727"],
["std","26.494628347919768","2.923430595455697","11.708825480998659","
245.88533510097207","234.17650961997307","6.131497611557303e-
14","11.708825480998659","1.718580294379123"],
["min","10.08","1.0","0.5085","10.6785","10.17","4.761904762","0.5085"
,"4.0"],
["25%","32.875","3.0","5.924875","124.422375","118.4975","4.761904762"
,"5.924875","5.5"],
["50%","55.230000000000004","5.0","12.088000000000001","253.848","241.
76","4.761904762","12.088000000000001","7.0"],
["75%","77.935","8.0","22.44525","471.35024999999996","448.90500000000
003","4.761904762","22.44525","8.5"],
["max","99.96","10.0","49.65","1042.65","993.0","4.761904762","49.65",
"10.0"]],"shape":{"columns":8,"rows":8}}

```

#Count of null values

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
```

```
RangeIndex: 1000 entries, 0 to 999
```

```
Data columns (total 17 columns):
```

#	Column	Non-Null Count	Dtype
0	Invoice ID	1000 non-null	object
1	Branch	1000 non-null	object
2	City	1000 non-null	object
3	Customer type	1000 non-null	object
4	Gender	1000 non-null	object
5	Product line	1000 non-null	object
6	Unit price	1000 non-null	float64
7	Quantity	1000 non-null	int64
8	Tax 5%	1000 non-null	float64
9	Total	1000 non-null	float64
10	Date	1000 non-null	object
11	Time	1000 non-null	object
12	Payment	1000 non-null	object
13	cogs	1000 non-null	float64
14	gross margin percentage	1000 non-null	float64
15	gross income	1000 non-null	float64
16	Rating	1000 non-null	float64

```
dtypes: float64(7), int64(1), object(9)
memory usage: 132.9+ KB
```

```
#Median
```

```
df.median(numeric_only=True)
```

```
{ "columns": [ { "name": "index", "rawType": "object", "type": "string" },
{ "name": "0", "rawType": "float64", "type": "float" } ], "ref": "63181f3d-d442-41bd-b0b8-dbb6ba2bcd07", "rows": [ [ "Unit price", "55.230000000000004" ],
[ "Quantity", "5.0" ], [ "Tax 5%", "12.088000000000001" ],
[ "Total", "253.848" ], [ "cogs", "241.76" ], [ "gross margin percentage", "4.761904762" ], [ "gross income", "12.088000000000001" ],
[ "Rating", "7.0" ] ], "shape": { "columns": 1, "rows": 8 } }
```

```
#Mode
```

```
print(df['Product line'].mode())
```

```
print(df['City'].mode())
```

```
print(df['Payment'].mode())
```

```
print(df['Customer type'].mode())
```

```
print(df['Gender'].mode())
```

```
0    Fashion accessories
```

```
Name: Product line, dtype: object
```

```
0    Yangon
```

```
Name: City, dtype: object
```

```
0    Ewallet
```

```
Name: Payment, dtype: object
```

```
0    Member
```

```
Name: Customer type, dtype: object
```

```
0    Female
```

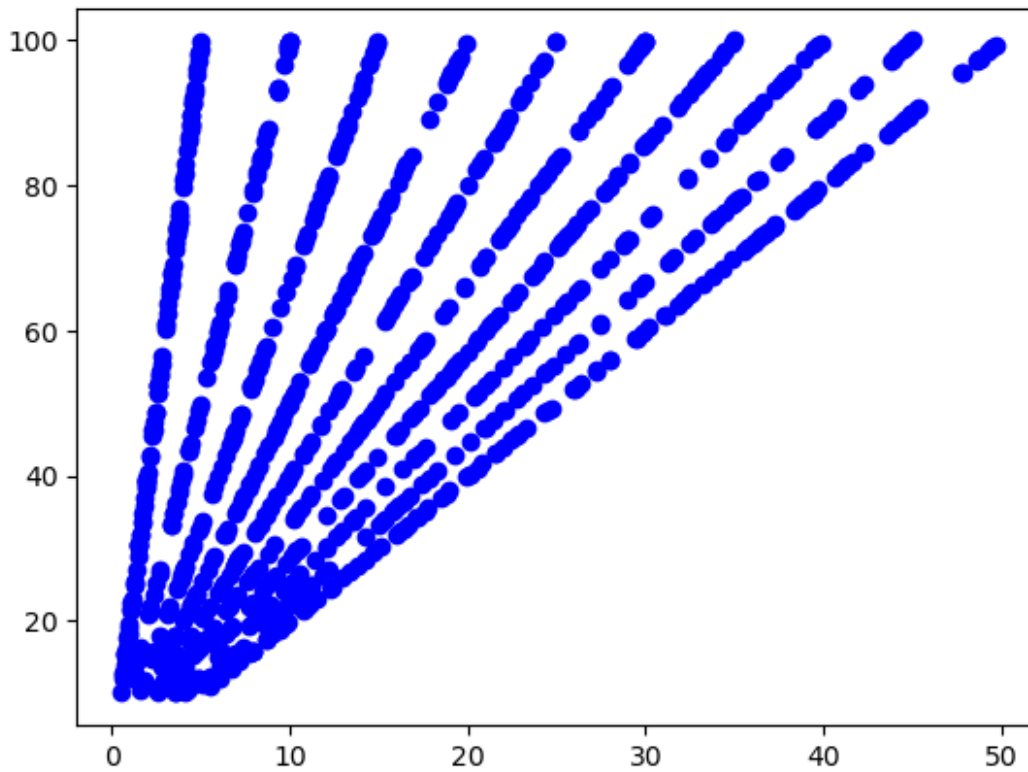
```
Name: Gender, dtype: object
```

```
#Scatter plot
```

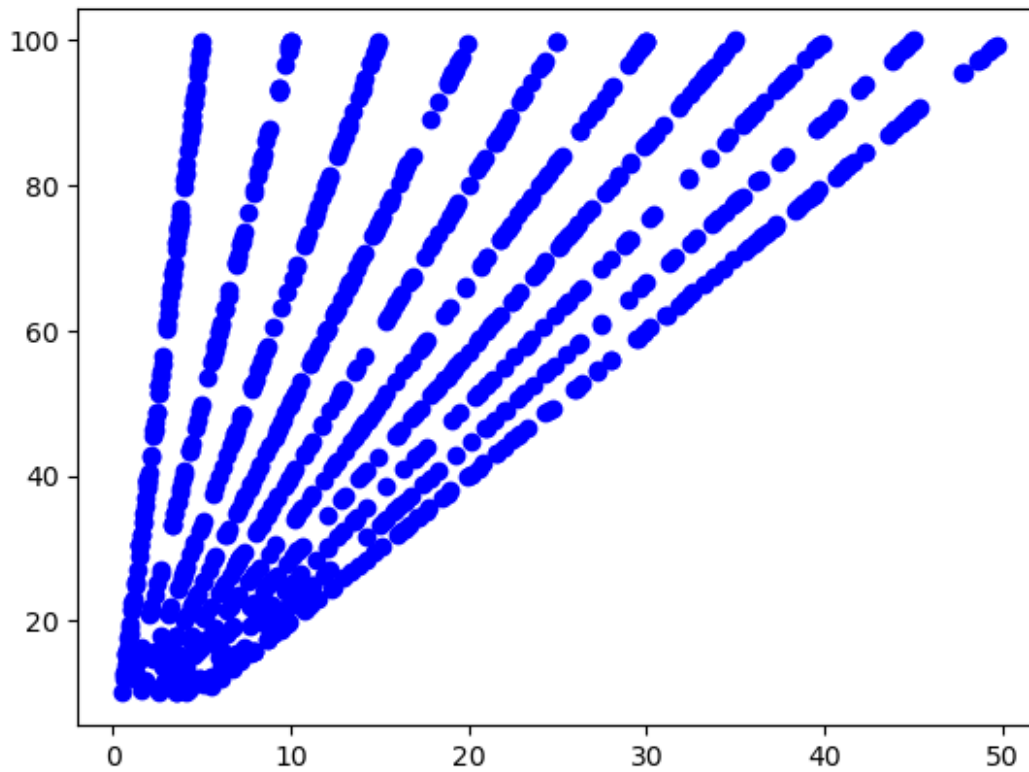
```
import matplotlib.pyplot as plt
```

```
plt.scatter(df['Tax 5%'], df['Unit price'], c = "blue")
```

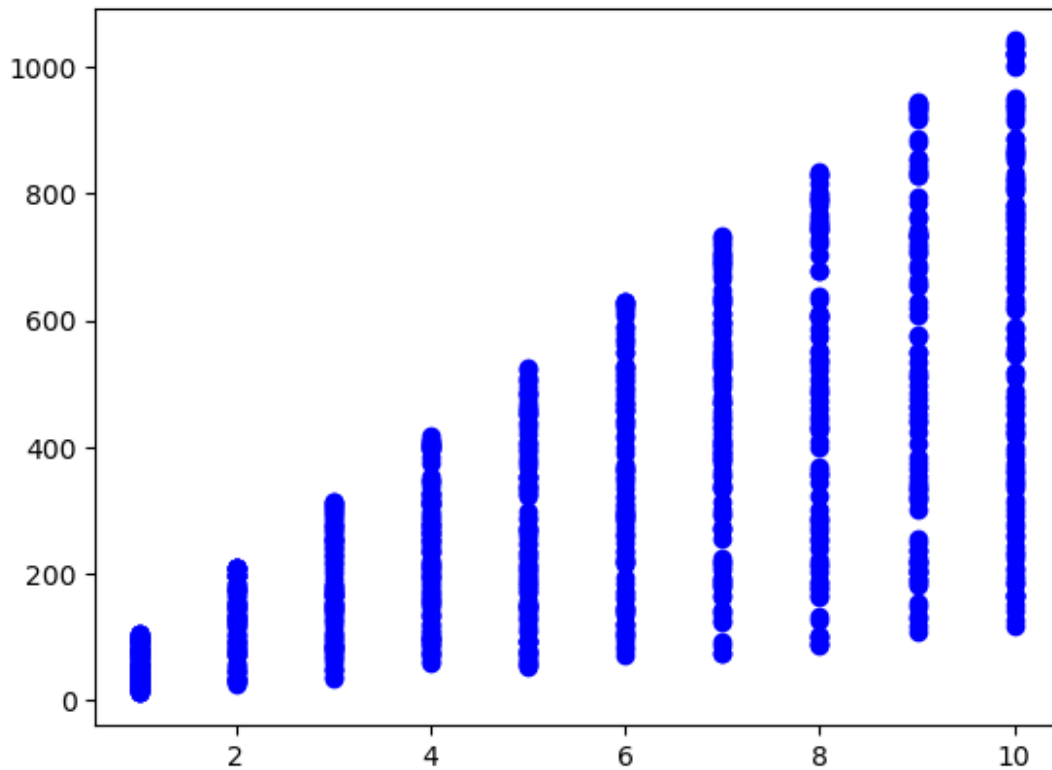
```
<matplotlib.collections.PathCollection at 0x1dd0ef702f0>
```



```
import matplotlib.pyplot as plt
plt.scatter(df['gross income'], df['Unit price'], c="blue")
<matplotlib.collections.PathCollection at 0x1dd0f7a60d0>
```

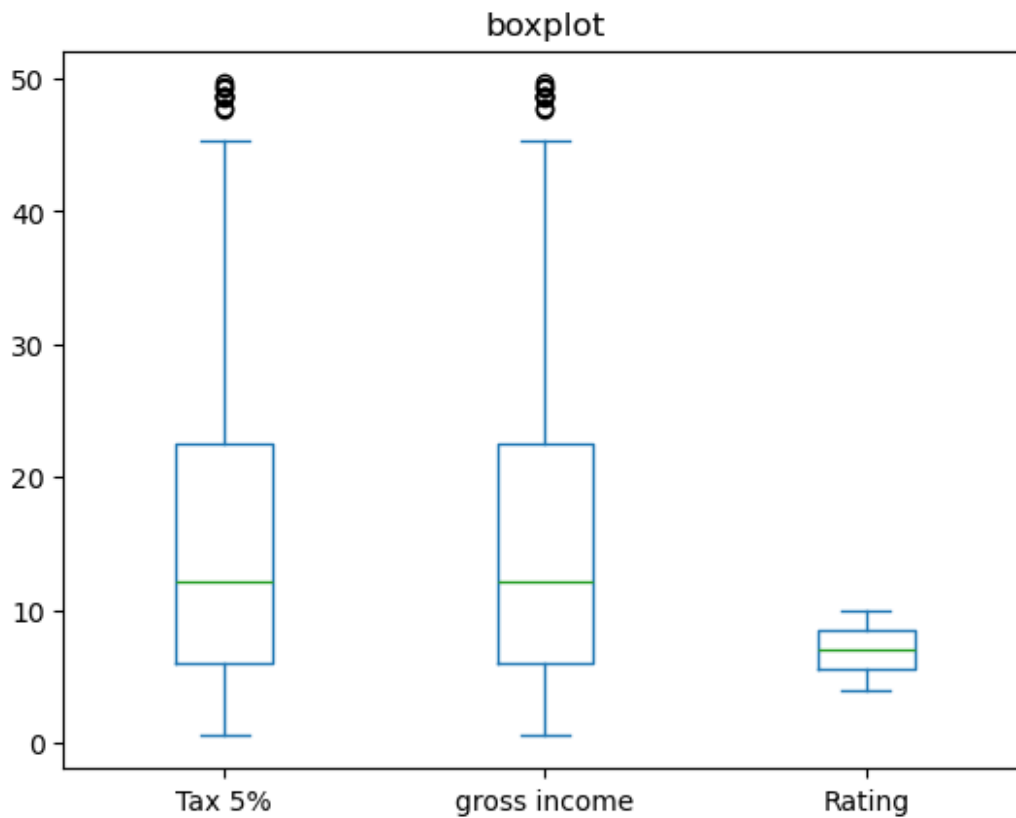


```
import matplotlib.pyplot as plt
plt.scatter(df['Quantity'], df['Total'], c="blue")
<matplotlib.collections.PathCollection at 0x1dd0f899d10>
```

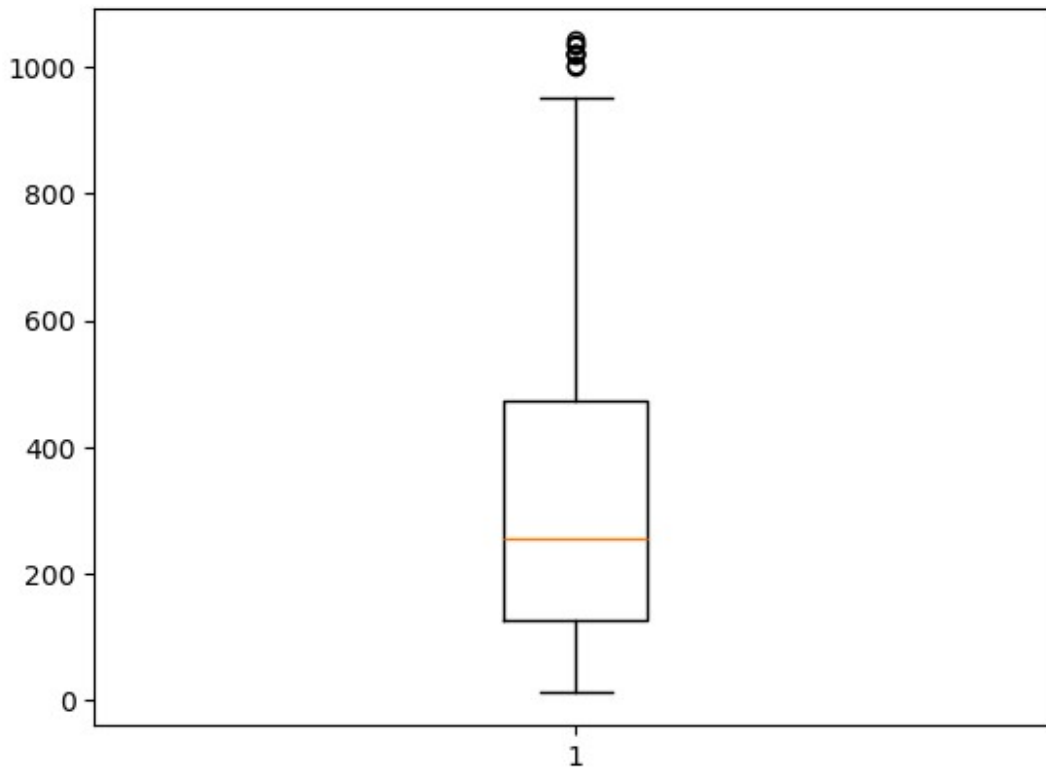


```
#Box plot
x2=df['Tax 5%']
x4=df['gross income']
x5=df['Rating']
data = pd.DataFrame({ "Tax 5%": x2,"gross income": x4,"Rating": x5})

# Plot the dataframe
ax = data[[ 'Tax 5%', 'gross income', 'Rating']].plot(kind='box',
title='boxplot')
```



```
plt.boxplot(df['Total'])  
{'whiskers': [<matplotlib.lines.Line2D at 0x1dd0f06a210>, <matplotlib.lines.Line2D at 0x1dd0f06a350>],  
'caps': [<matplotlib.lines.Line2D at 0x1dd0f06a490>, <matplotlib.lines.Line2D at 0x1dd0f06a5d0>],  
'boxes': [<matplotlib.lines.Line2D at 0x1dd0f06a0d0>],  
'medians': [<matplotlib.lines.Line2D at 0x1dd0f06a710>],  
'fliers': [<matplotlib.lines.Line2D at 0x1dd0f06a850>],  
'means': []}
```



```
#Trimmed mean
from scipy import stats
stats.trim_mean(df['Total'], 0.1)

np.float64(293.906615625)

#Summation
df['Total'].sum()

np.float64(322966.749)

#Frequency
count = df['Product line'].value_counts()
print(count)

Product line
Fashion accessories    178
Food and beverages    174
Electronic accessories 170
Sports and travel      166
Home and lifestyle     160
Health and beauty      152
Name: count, dtype: int64

#Variance
df.var(numeric_only=True)
```



```
{
  "columns": [
    {"name": "index", "rawType": "object", "type": "string"},
    {"name": "0", "rawType": "float64", "type": "float"}
  ],
  "ref": "a8fb450e-166b-4d51-8d2a-165c004ec307",
  "rows": [
    ["Unit price", "701.9653312943939"],
    ["Quantity", "8.546446446446451"],
    ["Tax 5%", "137.09659414448348"],
    ["Total", "60459.598017717326"],
    ["cogs", "54838.63765779334"],
    ["gross margin percentage", "3.75952629605329e-27"],
    ["gross income", "137.09659414448348"],
    ["Rating", "2.9535182282282335"]
  ],
  "shape": {"columns": 1, "rows": 8}
}
```

```
#Correlation matrix
df.corr(numeric_only=True)
```

```
{
  "columns": [
    {"name": "index", "rawType": "object", "type": "string"},
    {"name": "Unit price", "rawType": "float64", "type": "float"},
    {"name": "Quantity", "rawType": "float64", "type": "float"},
    {"name": "Tax 5%", "rawType": "float64", "type": "float"},
    {"name": "Total", "rawType": "float64", "type": "float"},
    {"name": "cogs", "rawType": "float64", "type": "float"},
    {"name": "gross margin percentage", "rawType": "float64", "type": "float"},
    {"name": "gross income", "rawType": "float64", "type": "float"},
    {"name": "Rating", "rawType": "float64", "type": "float"}
  ],
  "ref": "0b1ff78b-83fa-4a1f-8728-7646b75b0acf",
  "rows": [
    ["Unit price", "1.0", "0.010777564342497253", "0.6339620885890688", "0.6339620885890689", "0.633962088589068", null, "0.6339620885890688", "-0.008777507200778936"],
    ["Quantity", "0.010777564342497253", "1.0", "0.7055101859433074", "0.7055101859433058", "0.7055101859433045", null, "0.7055101859433074", "-0.015814904627167052"],
    ["Tax 5%", "0.6339620885890688", "0.7055101859433074", "1.0", "0.9999999999999989", "0.9999999999999991", null, "1.0", "-0.03644170499701844"],
    ["Total", "0.6339620885890689", "0.7055101859433058", "0.9999999999999989", "1.0", "0.9999999999999984", null, "0.9999999999999989", "-0.03644170499701849"],
    ["cogs", "0.633962088589068", "0.7055101859433045", "0.9999999999999991", "0.9999999999999984", "1.0", null, "0.999999999999999", "-0.03644170499701842"],
    ["gross margin percentage", null, null, null, null, null, null, null, null],
    ["gross income", "0.6339620885890688", "0.7055101859433074", "1.0", "0.9999999999999989", "0.999999999999999", null, "1.0", "-0.03644170499701844"],
    ["Rating", "-0.008777507200778936", "-0.015814904627167052", "-0.03644170499701844", "-0.03644170499701849", "-0.03644170499701842", null, "-0.03644170499701844", "1.0"]
  ],
  "shape": {"columns": 8, "rows": 8}
}
```

```
#Standard error of mean
df.sem(numeric_only=True)
```

```
{
  "columns": [
    {"name": "index", "rawType": "object", "type": "string"},
    {"name": "0", "rawType": "float64", "type": "float"}
  ],
  "ref": "bdd020b7-b344-4bd4-afde-9015cdccc939",
  "rows": [
    ["Unit price", "0.8378337133909057"],

```

```

["Quantity","0.09244699263062293"],["Tax 5%","0.3702655724537234"],
["Total","7.775577021528192"],["cogs","7.4053114490744685"],["gross
margin percentage","0.0"],["gross income","0.3702655724537234"],
["Rating","0.05434628072120693"]], "shape":{"columns":1,"rows":8}}

#sum of squares
sos=0
for val in df['Total']:
    sos=val*val+sos
print(sos)

164706659.37932873

#Skewness
df.skew(numeric_only=True)

{"columns":[{"name":"index","rawType":"object","type":"string"},
{"name":"0","rawType":"float64","type":"float"}],"ref":"749ca279-c5fc-
4089-975a-e9395cd11d74","rows":[["Unit price","0.007077447853328846"],
["Quantity","0.012941048017172435"],["Tax 5%","0.8925698049581423"],
["Total","0.8925698049581418"],["cogs","0.8925698049581418"],["gross
margin percentage","0.0"],["gross income","0.8925698049581423"],
["Rating","0.00900964876573073"]], "shape":{"columns":1,"rows":8}}

#kurtosis
sr = pd.Series(df['Total'])
print(sr.kurtosis())

-0.08188475790435135

import seaborn as sns

g=sns.displot(df['Total'])

```

